Toward a Gradient-Based Classification Space for Heterogeneous Complex Systems

An exploratory morphodynamic framework

Marie Maassouli

March 2026

Contents

1 Abstract **1**

2 Introduction **2**

3 Framework **3**
3.1 2.1 Morphological hypothesis . . . 3
3.2 2.2 Organizational axes . . . 4
3.2.1 A1 — Hierarchical depth . . . 4
3.2.2 A2 — Propagation capacity . . . 4
3.2.3 A3 — Integration . . . 4
3.2.4 A4 — Normativity . . . 4
3.2.5 A5 — Revision capacity . . . 5
3.3 2.3 Notation and gradients . . . 5

4 Gradient-Based Regime Classification **6**
4.1 3.1 Architecture . . . 6
4.2 3.2 Definition of regimes . . . 6
4.2.1 F01 — Differential equilibrium . . . 6
4.2.2 F03 — Propagation crisis (over-coupling) . . . 6
4.2.3 F04 — Inertial plateau . . . 6
4.2.4 F05 — Normative dominance (rigidification) . . . 6
4.2.5 F06 — Normative inversion (individuation) . . . 7
4.2.6 F08 — Functional dissolution . . . 7
4.2.7 F09 — Pure architecture . . . 7
4.3 3.3 Classification by constraint distance . . . 7
4.4 3.4 Regime overlaps . . . 7

5 Robustness Diagnostics **8**
5.1 4.1 Logical consistency (T1) — PASS . . . 8
5.2 4.2 Regime overlap and ambiguity (T2) — FAIL against conservative thresholds; structurally localized . . . 8
5.3 4.3 Coverage (T3) — FAIL under uniform-cube criterion; likely inflated by implausible configurations . . . 8

5.4 4.4 Local stability under noise (T4) — PARTIAL FAIL; boundary coherence high . . 9
5.5 4.5 Gradient invariance (T5) — PARTIAL PASS (T_1 passes; T_2, T_3 marginal) 9
5.6 4.6 Non-regression on the empirical case set (T6) — PASS 9

6 Empirical Study **9**
6.1 5.1 Case set 9
6.2 5.2 Distribution in gradient space 10
6.3 5.3 Key structural observations 11

7 Discussion **12**
7.1 6.1 Gradients as structural discriminants 12
7.2 6.2 The F01/F06 frontier and intrinsic dimensionality 12
7.3 6.3 Epistemic status 13
7.4 6.4 Limitations 14

8 Conclusion and Outlook **14**

9 Appendix A. Scoring Protocol **15**
9.1 A.1 General principle 15
9.2 A.2 A1 — Hierarchical depth 15
9.3 A.3 A2 — Propagation capacity (V2.1.1: scale-invariant formulation) 15
9.4 A.4 A3 — Integration 15
9.5 A.5 A4 — Normativity 15
9.6 A.6 A5 — Revision capacity (V2.1.1: endogenous plasticity formulation) 16

10 Appendix B. Case Table **16**

11 Appendix C. Diagnostic Summary **16**

12 Appendix D. Version History **17**

13 References **17**

1 Abstract

Understanding and comparing heterogeneous complex systems — biological, technological, institutional, and economic — remains difficult because most analytical frameworks are domain-specific. This work proposes an exploratory classification framework that maps such systems into a common property space defined by five observable organizational dimensions: hierarchical depth, propagation capacity, integration, normativity, and revision.

Rather than relying primarily on absolute values of these axes, the framework emphasizes structural gradients between them. In particular, three differences between axes (Δ_{23}: propagation–integration, Δ_{45}: normativity–revision, Δ_{12}: hierarchical depth–propagation) appear to provide the main discriminants of organizational regime structure. Systems are then classified into seven morphodynamic regimes defined as constraint-bounded regions of this multidimensional space.

An explicit scoring protocol based on cumulative sub-criteria is introduced to support reproducibility. A classifier implementing these definitions is evaluated through a battery of six diagnostic tests on 100,000 uniformly sampled points, including logical consistency checks, overlap analysis, coverage tests of the unit hypercube, local stability under noise, invariance under axis

translations that preserve gradients, and non-regression on a set of fifteen heterogeneous case studies.

Results indicate that the gradient Δ_{23} (propagation–integration) is the strongest regime discriminant, with an invariance rate of 97.5% under gradient-preserving transformations. Regime boundaries are found to be localized rather than diffuse, with the principal ambiguity concentrated on the equilibrium–individuation frontier (F01/F06). The classifier achieves 15/15 correct classifications on the empirical battery with no regression from earlier versions.

The framework should be understood as an exploratory instrument rather than a completed theory. Its primary contribution is to suggest that gradient relationships between organizational properties may provide a useful basis for cross-domain classification of complex systems. Limitations include the use of a single annotator, a small initial case set, and empirically determined regime thresholds.

Code and data availability. The classifier, scoring protocol, diagnostic scripts, and figure generation code are available at `https://github.com/MMli97/1-morphodynamic-index`.

2 Introduction

Understanding and comparing heterogeneous complex systems remains a persistent challenge across scientific disciplines. Biological organisms, technological infrastructures, social institutions, and economic networks all exhibit complex forms of organization, yet the conceptual frameworks used to analyze them are often domain-specific. As a result, comparative analysis across domains typically relies on metaphor or analogy rather than on shared formal instruments.

The ambition to identify structural principles common to diverse systems has a long history. Von Bertalanffy (1968) proposed the existence of structural isomorphisms between systems of different natures, arguing that the same differential equations could describe population growth, market expansion, and substance diffusion. His *General System Theory* articulated the need for a transdisciplinary framework, but the program remained largely programmatic: it identified formal analogies without producing an instrumentable common space in which concrete systems could be positioned and compared.

Cybernetic approaches introduced more precise models of regulation and feedback. Ashby (1956) formulated the law of requisite variety, establishing that a system can only regulate what it can distinguish — a principle that anticipates the concept of internal normativity used in the present framework. Beer (1972) extended these ideas into organizational theory with his Viable System Model, decomposing viable organizations into five recursive subsystems. However, the VSM remains prescriptive and qualitative: it describes how a system *should* be organized for viability, but does not provide a metric space for comparing how systems *are* organized.

In the philosophy of biology, Canguilhem (1966) introduced a distinction that proves central to the present work: between maintaining a fixed norm (the *normal*) and the capacity to produce new norms in response to changed conditions (the *normative*). This distinction between regulation and meta-regulation — between applying rules and revising rules — is formalized here as the independence of axes A_4 (normativity) and A_5 (revision). Maturana and Varela (1980) extended this line with the concept of autopoiesis, emphasizing that living systems produce themselves while maintaining organizational closure. Their notion of structural coupling informs the definition of axis A_2 (propagation), while their insistence on internal normativity shapes the distinction between endogenous and imposed regulation in A_4.

From information theory, Tononi (2004) proposed the integrated information measure Φ, quantifying the extent to which a system generates information beyond the sum of its parts. This

measure is the direct ancestor of axis A_3 (integration). However, Φ is a scalar: it indicates *how much* a system integrates, but not *in what structural relationship* integration stands to other organizational properties such as propagation or normativity. The present framework extends this intuition to a multidimensional space where integration is one axis among several, and where the gradient $\Delta_{23} = A_2 - A_3$ captures the structural tension between propagation and integration.

In ecology, Holling (1973, 2001) developed the adaptive cycle framework, identifying four phases of systemic dynamics: exploitation, conservation, release, and reorganization. Walker and Salt (2006) extended this into the concept of panarchy. The correspondence with the present framework is nearly direct: the conservation phase maps to equilibrium (F01), the release phase to propagation crisis (F03), and reorganization to normative inversion (F06). What the present framework adds is a metric formalization: rather than describing phases qualitatively, it positions systems in a gradient space where transitions correspond to threshold crossings along Δ_{23} and Δ_{45}.

Kauffman (1993) showed that complex systems self-organize near the "edge of chaos," a regime that is neither fully ordered nor fully disordered. Within the present framework, equilibrium (F01) corresponds to this edge: the gradients are near zero, and the system neither over-couples nor rigidifies. What the gradient-based approach adds is the ability to distinguish *how* a system leaves the edge — by over-coupling (Δ_{23} increasing toward F03), by rigidification (Δ_{45} increasing toward F05), or by normative collapse (A_4, A_5 decreasing toward F09).

Despite these contributions, no existing framework provides what might be called a *shared instrumentable projection space*: a device in which a eukaryotic cell, a financial market, and a large language model can be positioned on the same axes, and where positions are not metaphors but measurements — approximate, revisable, and formalized.

The present work explores this possibility. We introduce a five-dimensional organizational space, define seven regimes as constraint-bounded regions within this space, and evaluate the resulting classifier through a battery of diagnostic tests. The goal is methodological: rather than proposing a general theory of complex systems, we introduce a gradient-based classification instrument intended as a starting point for systematic cross-domain comparison.

The approach is consistent with the epistemological perspective articulated by van Fraassen (2008), in which scientific models are evaluated by their capacity to provide empirically adequate representations of phenomena rather than by their correspondence to underlying ontological structures. The framework proposed here functions as a representational construction — a projection apparatus, not a mirror of reality.

The methodology of iterative instrument correction adopted throughout this work follows the pattern identified by Chang (2004) in his study of temperature measurement. Chang showed that the construction of the thermometer involved a century of iterative bias correction without questioning the existence of temperature itself. The trajectory of the present classifier (V1 → V2 → V2.1.2) reproduces this logic at a compressed scale: biases are identified and corrected in the instrument while the theoretical framework remains stable.

This article is a method paper presenting an exploratory framework, not a validation study establishing definitive results. Its contribution is the framework itself — the axes, gradients, regimes, classifier, and diagnostic protocol — offered as an explicit and testable starting point for further investigation.

The remainder of the article is organized as follows. Section 2 introduces the organizational framework and defines the axes, scoring protocol, and formal notation. Section 3 presents the gradient-based regime classification and the associated classifier. Section 4 describes the diagnostic tests and reports their results. Section 5 presents the empirical case study. Section 6 discusses the implications and limitations, and Section 7 concludes with directions for future work.

3 Framework

3.1 2.1 Morphological hypothesis

The framework proposed here is based on the hypothesis that heterogeneous complex systems can be represented within a common descriptive space defined by a small number of organizational properties. These properties do not aim to capture the internal mechanisms of each system but rather structural features of how the system organizes propagation, integration, and regulation of processes.

In this perspective, a system is characterized not by its domain (biological, technical, social) but by its position within a space of organizational properties. Differences between systems can therefore be analyzed geometrically as differences in the configuration of these properties.

The framework does not assume that these properties correspond to fundamental ontological dimensions. They are introduced as operational descriptors intended to support comparative analysis. This position follows the semantic view of scientific theories (Giere, 1988; van Fraassen, 1980), in which a theory is identified through its class of models and the relationship between those models and empirical systems, rather than through a set of axiomatic propositions.

3.2 2.2 Organizational axes

Each system is represented by a vector

$$x = (A_1, A_2, A_3, A_4, A_5) \in [0,1]^5$$

where each component corresponds to a normalized organizational axis. The axes are observable descriptors, not fundamental variables: they are chosen for operational measurability across domains, not claimed to exhaust the properties of complex systems. The gradients defined in Section 2.3 are the principal empirically observed discriminants of regime structure.

The five axes were selected to satisfy five admissibility criteria: continuous gradation on [0,1], demonstrated independence on discriminating cases, applicability across domains, non-normativity (the axis does not prescribe what is "good"), and the existence of conceivable formal proxies.

3.2.1 A1 — Hierarchical depth

Hierarchical depth describes the number and differentiation of causal layers within the system. This is not system size or element count, but stratification: how many levels of transformation must causality traverse to produce the global state. Bar-Yam (1997) showed that multi-scale complexity is not reducible to element count; Ladyman, Lambert, and Wiesner (2013) reviewed complexity definitions and found none to be universal. Hierarchical depth was chosen as the most operationally measurable and least ambiguous among available candidates.

3.2.2 A2 — Propagation capacity

Propagation capacity measures the global sensitivity of a system to local perturbation. If a local change δe produces a significant global change ΔS, propagation is high. This corresponds to the notion of structural coupling in Maturana and Varela (1980), restricted here to perturbation dynamics. The 2010 Flash Crash illustrates maximal propagation; the modular architecture of Linux attenuates it.

3.2.3 A3 — Integration

Integration measures the reduction of global variance produced by internal coordination. This is the most direct descendant of Tononi's Φ (2004), but the distinction with A_2 is crucial: a system can propagate strongly (A_2 high) without stabilizing (A_3 low). A speculative bubble exhibits converging correlations (high propagation) while price variance explodes (collapsed integration). Recent work by Mediano, Rosas, and colleagues (2022) on information decomposition offers promising formal proxies for empirical measurement of A_3.

3.2.4 A4 — Normativity

Normativity describes the capacity of a system to restrict its state space by producing and maintaining a preferred region $R \subset \Omega$ through internal correction mechanisms. This definition draws on Ashby's requisite variety (1956), with the precision introduced by Canguilhem (1966): only normativity *produced* by the system counts, not normativity *imposed* from outside. A thermostat has a simple attractor (its setpoint). An authoritarian regime has an imposed attractor. An LLM during inference has none ($A_4 = 0$).

3.2.5 A5 — Revision capacity

Revision capacity describes the ability of a system to modify its own regulatory mechanisms (D) or the criteria defining its admissible region (R). A_4 applies rules; A_5 modifies rules. This distinction formalizes Canguilhem's contrast between *le normal* and *le normatif*. Simondon (1958/2005) conceptualized individuation as a continuous process of reorganization; regime F06 (normative inversion, where $A_5 > A_4$) is the formal translation of this Simondonian intuition. Varela, Thompson, and Rosch (1991) extended this to biological autonomy, arguing that structural plasticity — chromatin remodeling, synaptic plasticity, signaling network reconfiguration — constitutes revision in the functional sense.

Each axis is operationalized through five cumulative sub-criteria scored 0, 0.5, or 1. The normalized score corresponds to the mean of satisfied criteria. The complete scoring protocol, including the exact sub-criteria for each axis, is provided in Appendix A. This protocol aims to support inter-annotator reproducibility, although calibration across annotators remains future work.

3.3 2.3 Notation and gradients

While the framework is defined on five axes, empirical exploration suggests that relationships between axes play a central role in distinguishing organizational regimes. We therefore introduce three primary gradients:

$$\Delta_{23} = A_2 - A_3 \quad \text{(propagation vs. integration)}$$

$$\Delta_{45} = A_4 - A_5 \quad \text{(normativity vs. revision)}$$

$$\Delta_{12} = A_1 - A_2 \quad \text{(hierarchical depth vs. propagation)}$$

Two secondary gradients are also used in specific regime definitions:

$$\Delta_{35} = A_3 - A_5 \quad \text{(integration vs. revision)}$$

$$\Delta_{43} = A_4 - A_3 \quad \text{(normativity vs. integration)}$$

The primacy of gradients over absolute values as regime discriminants was not assumed *a priori* but emerged from the analysis of the initial case set: an early version (V1) defining regimes by absolute thresholds achieved only 3/15 correct classifications, whereas a gradient-primary architecture (V2) achieved 15/15.

This observation has a conceptual antecedent in catastrophe theory. Thom (1972) showed that morphological forms are determined not by absolute parameter values but by relationships between parameters; qualitative changes (catastrophes) occur where these relationships cross critical surfaces. The transition from V1 to V2 in the present framework is structurally analogous: regimes are defined by gradient relations, and transitions between regimes correspond to threshold crossings along these gradients.

4 Gradient-Based Regime Classification

4.1 3.1 Architecture

The classifier operates on two levels:

- **Level 1 — Gradient typing.** Regimes are defined primarily by conditions on gradients Δ_{23}, Δ_{45}, Δ_{12}, Δ_{35}, Δ_{43}. Gradient constraints carry weights of 2.0–2.5 in the distance computation.
- **Level 2 — Absolute qualification.** Absolute axis values qualify the intensity of the regime. Absolute constraints carry weights of 0.8–1.5.

This two-level architecture reflects the empirical finding that regime *type* depends on inter-axis relationships, while regime *intensity* depends on absolute levels.

4.2 3.2 Definition of regimes

Seven regimes are defined in the current version (V2.1.2). Each regime corresponds to a constraint-bounded region of the organizational space.

4.2.1 F01 — Differential equilibrium

Propagation and integration are synchronized; normativity and revision are balanced. The system absorbs perturbations without over-coupling and regulates without locking out revision.

Primary constraints: $\Delta_{23} \leq 0.18$, $\Delta_{45} \leq 0.30$. Secondary: $A_3 \geq 0.40$, $A_4 \geq 0.25$, $A_5 \geq 0.15$.

This is the most populated regime (7/15 cases), corresponding to Walker and Salt's (2006) "resilience basin."

4.2.2 F03 — Propagation crisis (over-coupling)

Propagation exceeds integration. Perturbations diffuse faster than the system stabilizes them. In Holling's (1973) terminology, this is the "release" phase where accumulated connections become cascade vectors.

Primary: $\Delta_{23} \geq 0.23$. Secondary: $A_4 \leq 0.50$, $A_2 \geq 0.55$.

4.2.3 F04 — Inertial plateau

Integration dominates revision ($\Delta_{35} \geq 0.35$) and depth dominates propagation ($\Delta_{12} \geq 0.15$). The system is structurally complete but non-revisable.

Secondary: $A_1 \geq 0.60$, $A_5 \leq 0.30$.

4.2.4 F05 — Normative dominance (rigidification)

Normativity strongly exceeds revision. The system regulates without ever revising its criteria. In Canguilhem's (1966) terms, the system has become *normal* without remaining *normative*.

Primary: $\Delta_{45} \geq 0.40$. Secondary: $A_4 \geq 0.45$, $A_5 \leq 0.30$.

4.2.5 F06 — Normative inversion (individuation)

Revision equals or exceeds normativity. The system reorganizes actively, modifying its own rules. This is the Simondonian moment: active individuation, the production of new forms. Propagation is stabilized ($\Delta_{23} \leq 0.18$).

Primary: $\Delta_{45} \leq 0.05$. Secondary: $A_5 \geq 0.50$, $A_3 \geq 0.45$.

4.2.6 F08 — Functional dissolution

Normativity exceeds integration ($\Delta_{43} \geq 0.25$) and depth exceeds propagation ($\Delta_{12} \geq 0.20$). Norms persist but no longer stabilize; structure exists but no longer propagates.

Secondary: $A_3 \leq 0.40$, $A_5 \leq 0.20$.

4.2.7 F09 — Pure architecture

Integration vastly exceeds normativity ($\Delta_{43} \leq -0.40$) with near-zero normativity ($A_4 \leq 0.15$) and revision ($A_5 \leq 0.10$). The system integrates at scale without any internal correction mechanism. This regime is introduced as a limit configuration for systems exhibiting strong integration and complexity with minimal normativity and revision. Static large language models during inference are one illustrative example, but other candidates include large databases, crystal lattices, or infrastructure networks without feedback control.[1]

4.3 3.3 Classification by constraint distance

For each regime F, defined by a set of weighted inequalities $g_j(x) \leq 0$ with weights w_j, the classifier computes:

$$d_{\mathrm{F}}(x) = \sum_j w_j \cdot \max(0, g_j(x))$$

The classifier assigns x to the regime minimizing this distance:

$$\hat{\mathrm{F}}(x) = \arg\min_{\mathrm{F}} d_{\mathrm{F}}(x)$$

[1] The regime numbering (F01, F03, F04, F05, F06, F08, F09) preserves the development history of the framework. Regimes F02 and F07, which appeared in early exploratory versions, were removed during formalization because they did not satisfy the admissibility criteria or lacked discriminating cases. The numbering is retained for traceability.

The classification margin $m(x) = d_{\text{second}}(x) - d_{\text{best}}(x)$ provides a local indicator of confidence. The classifier is fully explainable: for any system, each constraint can be identified as satisfied or violated, with its violation magnitude and weight.

Figure 1: Classifier pipeline: input vectors are mapped to gradients, evaluated against regime constraints, and assigned to the nearest regime by weighted violation distance.

4.4 3.4 Regime overlaps

The classifier admits explicit overlaps between regimes. Systems may simultaneously satisfy the constraints of multiple regimes. These overlaps are not defects — they are structurally informative transition zones.

Two corridors are consistently observed:

- **Rigid corridor** (F04 ∩ F05): systems that are both saturated and normatively dominant.
- **Living corridor** (F01 ∩ F06): systems in equilibrium that maintain active revision capacity.

5 Robustness Diagnostics

The classifier was evaluated through six diagnostic tests on 100,000 uniformly sampled points in $[0,1]^5$. The tests assess logical consistency, regime separation, spatial coverage, local stability, gradient invariance, and empirical non-regression.

5.1 4.1 Logical consistency (T1) — PASS

Each regime was verified for satisfiability: all seven admit non-empty feasible regions in $[0,1]^5$. No contradictory constraint pairs were detected. Admissible points are often located near the boundaries of the unit hypercube, which is expected for an inequality-based classifier.

5.2 4.2 Regime overlap and ambiguity (T2) — FAIL against conservative thresholds; structurally localized

For each sampled point, the classification margin $m(x) = d_{\text{second}} - d_{\text{best}}$ was computed.

Metric	Value
$P(m < 0.01)$	19.7%
$P(m < 0.02)$	22.0%
Median m	0.149

The dominant ambiguity source is the F01/F06 frontier (21% of low-margin cases). This boundary is structurally soft because both regimes share the condition $\Delta_{23} \leq 0.18$ and differ primarily on Δ_{45} and A_5. The documented overlaps (F04 ∩ F05 at 2.6%, F01 ∩ F04 at 1.6%) are expected corridor behaviors. The F01/F03 grey zone, reshaped in V2.1.2, accounts for only 0.8% of ambiguities.

5.3 4.3 Coverage (T3) — FAIL under uniform-cube criterion; likely inflated by implausible configurations

Metric	Value
$P(d_{\min} > 0.10)$	24.6%
$P(d_{\min} > 0.15)$	17.9%
Inside rate ($d = 0$)	58.4%

F01 captures 42% of the unit cube volume but with a median margin of only 0.050, indicating soft dominance. Dead zones (high $d_{\min}$) correspond primarily to implausible system configurations (e.g., very low depth with very high propagation and normativity). The 15 empirical cases all fall at $d = 0$, suggesting that the dead zones are artifacts of uniform sampling rather than gaps in the classifier's coverage of real systems.

5.4 4.4 Local stability under noise (T4) — PARTIAL FAIL; boundary coherence high

Perturbation tests at $\sigma = 0.02$ and $\sigma = 0.05$:

Metric ($\sigma = 0.02$)	Value
Confident points ($m \geq 0.05$): proportion with ≥ 1 flip	5.6%
Boundary points oscillating ≤ 2 forms	97.7%

The convention used here counts the *proportion of confident points experiencing at least one regime change* across 20 perturbation draws. This is a conservative criterion: the *mean* flip rate per draw is 0.8%, well below 2%, but the per-point criterion yields 5.6%, exceeding the 2% target. The exceedance is driven by the soft F01/F06 frontier. Boundary oscillation remains confined to two-form oscillation in 97.7% of cases, indicating local coherence even in ambiguous zones.

5.5 4.5 Gradient invariance (T5) — PARTIAL PASS (T_1 passes; T_2, T_3 marginal)

Three gradient-preserving transformations were tested:

Transform	Gradient preserved	Regime preservation rate
T_1: $A_2 \mathrel{+}= t, A_3 \mathrel{+}= t$	Δ_{23}	97.5%
T_2: $A_4 \mathrel{+}= t, A_5 \mathrel{+}= t$	Δ_{45}	93.4%
T_3: $A_1 \mathrel{+}= t, A_2 \mathrel{+}= t$	Δ_{12}	93.8%

T_1 passes cleanly, confirming that Δ_{23} is the primary regime discriminant. T_2 and T_3 fall marginally below the 95% target due to absolute constraints at Level 2 and correlations between gradients (T_3 preserves Δ_{12} but shifts Δ_{23}). This is a structural consequence of the two-level architecture, not a defect of the gradient hypothesis.

5.6 4.6 Non-regression on the empirical case set (T6) — PASS

All fifteen systems are classified inside their expected regimes ($d = 0$) with correct primary assignments. Six cases exhibit zero margin, confirming their location on documented overlap boundaries (F04/F05, F04/F09, F01/F06).

6 Empirical Study

6.1 5.1 Case set

The case set comprises fifteen systems selected by maximal contrast, not representativeness. Each system was chosen to stress-test at least one axis or gradient:

- **Biological:** eukaryotic cell, multicellular organism, adaptive immune system
- **Institutional:** Fifth French Republic, authoritarian regime, Catholic Church, Reformed religious order
- **Technological:** thermostat, static large language model, TikTok platform
- **Economic:** NYSE financial market, global supply chain
- **Organizational:** Apple Inc., Linux development ecosystem
- **Disciplinary:** theoretical physics community

The complete scoring table is provided in Appendix B.

6.2 5.2 Distribution in gradient space

Figure 2 projects the fifteen systems onto the $(\Delta_{23}, \Delta_{45})$ plane. Several structural patterns emerge:

- **Rigid corridor** (upper region, $\Delta_{45} > 0.40$): Church (+0.65), authoritarian regime (+0.75), Reformed order (+0.45), thermostat (+0.50). Systems that regulate without revising.
- **Living corridor** (lower-left, $\Delta_{45} \approx 0$, $\Delta_{23} < 0$): Linux (+0.05), theoretical physics (−0.05). Systems in equilibrium with active revision.
- **Crisis region** (right, $\Delta_{23} > 0.20$): supply chain (+0.35), NYSE (+0.25). Propagation exceeds integration.
- **Limit configuration**: the static LLM ($\Delta_{43} = -0.70$) occupies a distinctive position — strong integration without normativity. This type of configuration appears not to have been explicitly conceptualized in earlier systems-theoretic frameworks, though it may find partial antecedents in discussions of passive information-processing architectures.

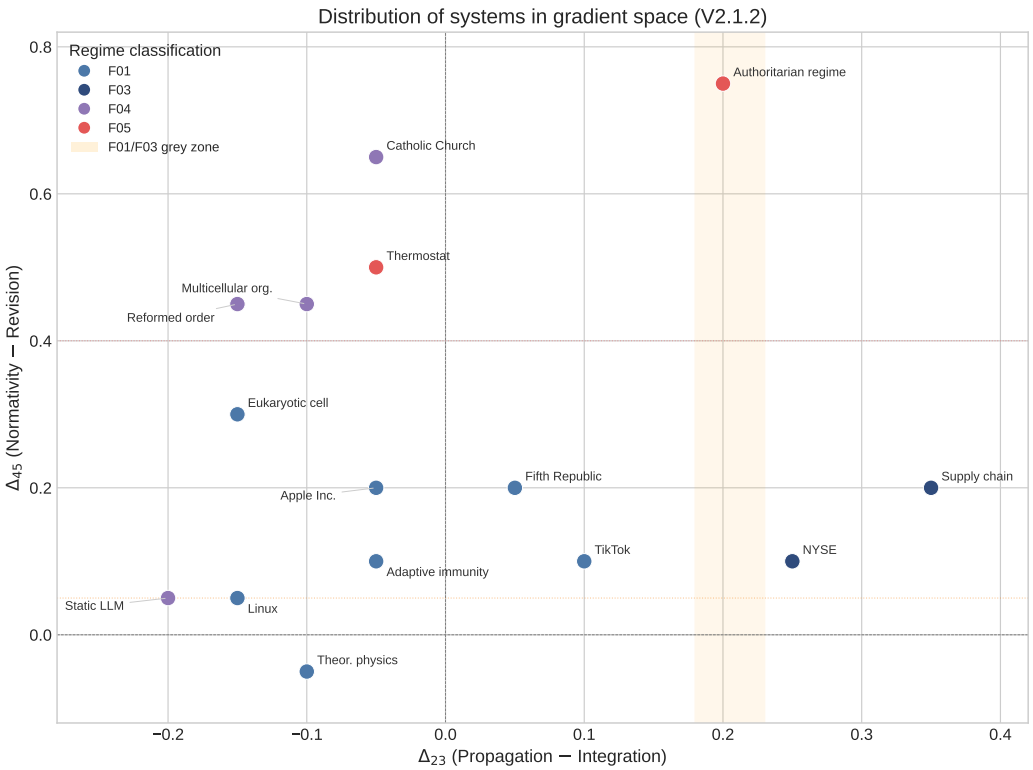

Figure 2: Distribution of fifteen systems in the $(\Delta_{23}, \Delta_{45})$ gradient plane. Colors indicate primary regime assignment. The shaded band marks the F01/F03 grey zone ($\Delta_{23} \in [0.18, 0.23]$). Dashed horizontal lines indicate the F05 threshold ($\Delta_{45} = 0.40$) and the F06 threshold ($\Delta_{45} = 0.05$).

6.3 5.3 Key structural observations

The gradient Δ_{23} is the most powerful discriminant. It cleanly separates crisis systems (supply chain +0.35, NYSE +0.25) from equilibrium systems (cell −0.15, Linux −0.15). The grey zone $[0.18, 0.23]$ is narrow.

The A_4/A_5 independence is demonstrated empirically: a thermostat has $A_4 = 0.50$, $A_5 = 0.00$ (high normativity, zero revision), while theoretical physics has $A_4 = 0.55$, $A_5 = 0.60$ (moderate normativity, maximal revision). This proves that regulation and meta-regulation are structurally independent.

Thermostat. The thermostat ($A_4 = 0.50$, $A_5 = 0.00$, $\Delta_{45} = +0.50$) serves as the minimal demonstration of A_4/A_5 independence. It regulates perfectly but never revises — its classification as F05 (normative dominance) is unambiguous and provides a structural anchor for the normativity–revision axis.

Catholic Church. The Church ($\Delta_{45} = +0.65$, $\Delta_{12} = +0.25$) sits squarely in the F04 $\cap$ F05 overlap — the "rigid corridor." It is both structurally saturated (inertial plateau: deep hierarchy without propagation) and normatively dominant (regulation without revision). This dual classification is not an artifact but an informative structural diagnosis: rigidification on a saturated plateau.

Linux ecosystem. Linux ($\Delta_{45} = +0.05$, $\Delta_{23} = -0.15$) occupies the F01 $\cap$ F06 overlap — the "living corridor." It maintains equilibrium while actively revising its own organizational rules. This combination of stability and meta-regulatory capacity places it at the intersection of two

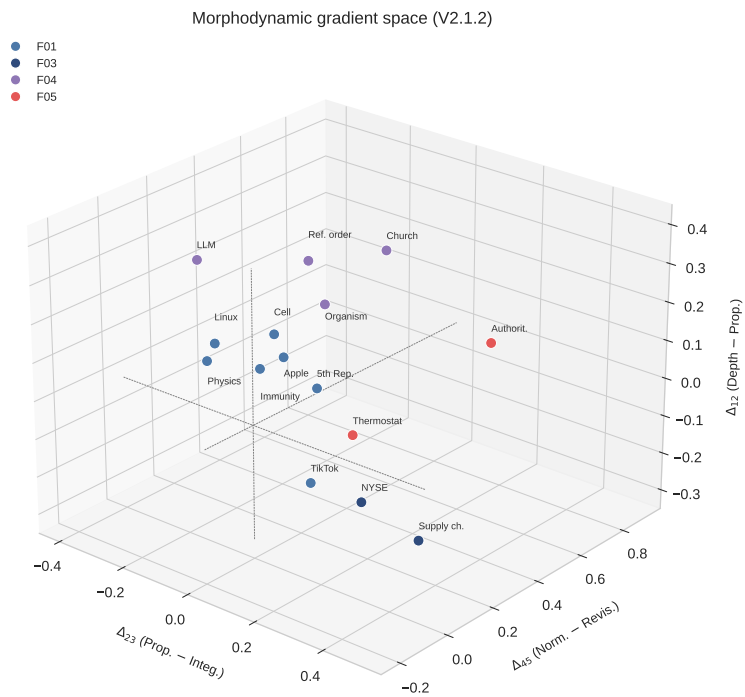

Figure 3: Three-dimensional projection of the fifteen systems in gradient space ($\Delta_{23}, \Delta_{45}, \Delta_{12}$). The vertical axis ($\Delta_{12}$) separates inertial systems (positive: Church, LLM) from diffusive systems (negative: Supply chain, TikTok).

regimes, illustrating how the framework captures systems that are simultaneously homeostatic and self-transforming.

Static LLM. The static LLM ($A_4 = 0.05$, $A_5 = 0.00$, $\Delta_{43} = -0.70$) is the most structurally distinctive case in the battery. It integrates information at scale without any normativity or revision, placing it in the F04 $\cap$ F09 overlap. This limit configuration illustrates the framework's capacity to represent organizational structures that differ fundamentally from biological or institutional systems.

Documented overlaps correspond to interpretable structural corridors. The Holling adaptive cycle (exploitation $\rightarrow$ release $\rightarrow$ reorganization $\rightarrow$ conservation) admits a structural correspondence with the transition sequence F01 $\rightarrow$ F03 $\rightarrow$ F06 $\rightarrow$ F01 in the gradient space, although the mapping is approximate and the frameworks differ in their formal commitments.

Complete scores, gradients, and regime assignments for all fifteen systems are provided in Table 1 (Appendix B).

7 Discussion

7.1 6.1 Gradients as structural discriminants

The most notable finding is the primacy of gradient relationships over absolute axis values for regime discrimination. This was not assumed *a priori* but emerged from the failure of V1 (absolute thresholds, 3/15 inside) and the success of V2+ (gradient-primary, 15/15 inside). The invariance

test (T5) provides independent confirmation: Δ_{23} is preserved at 97.5% under gradient-preserving transformations.

This observation resonates with Thom's (1972) insight that morphological forms depend on parameter *relations* rather than parameter *values*. It also echoes findings in other complex systems domains where intrinsic dimensionality is substantially lower than observational dimensionality (Bengio et al., 2013; the manifold hypothesis in representation learning).

7.2 6.2 The F01/F06 frontier and intrinsic dimensionality

The principal structural weakness of the current classifier is the soft boundary between equilibrium (F01) and normative inversion (F06). These two regimes share the condition $\Delta_{23} \leq 0.18$ and differ primarily on Δ_{45} (≤ 0.30 for F01, ≤ 0.05 for F06) and A_5 (≥ 0.15 for F01, ≥ 0.50 for F06). The wide band $\Delta_{45} \in [0.05, 0.30]$ combined with $A_5 \in [0.15, 0.50]$ creates a region where F01 wins by default with near-zero margin.

This may reflect a genuine structural feature rather than a classification defect: systems transitioning from equilibrium to active individuation may pass through this zone gradually rather than abruptly. The documented overlap (F01 $\cap$ F06 for Linux and theoretical physics) supports this interpretation.

More broadly, the diagnostic results suggest that the uniform cube $[0,1]^5$ substantially overestimates the effective space of plausible system configurations. Dead zones (24.6% of the cube at $d_{\min} > 0.10$) correspond to configurations that are structurally implausible — for example, minimal hierarchical depth combined with maximal normativity and revision. The fifteen empirical cases all fall at $d = 0$, and the meaningful regime structure appears to be concentrated on a lower-dimensional subset of the cube. This is consistent with the hypothesis that the intrinsic dimensionality of the organizational space may be closer to three (the gradient space $\Delta_{23}, \Delta_{45}, \Delta_{12}$) than to five.

7.3 6.3 Epistemic status

The framework functions as a representational construction in the sense of van Fraassen (2008): a projection apparatus, not a mirror of reality. The regimes are not claimed to be "real" attractors in a dynamical sense — no equation of evolution is defined on $[0,1]^5$. They are regions of a property space whose empirical adequacy is evaluated through diagnostic tests and case studies.

The methodology follows Chang's (2004) model of iterative instrument correction: biases are identified and corrected in the measurement instrument (the scoring protocol and classifier thresholds) without modifying the theoretical framework (the five axes and the gradient hypothesis). The trajectory V1 $\rightarrow$ V2 $\rightarrow$ V2.1.1 $\rightarrow$ V2.1.2 illustrates this process: two scoring corrections (scale invariance for A_2, endogenous plasticity for A_5) and one classifier adjustment (F01/F03 boundary) were applied, while the axis definitions and the gradient-primary architecture remained stable throughout.

7.4 6.4 Limitations

Several limitations must be emphasized:

1. **Single annotator.** The scoring protocol was applied by a single annotator. Although the cumulative sub-criteria design supports reproducibility, inter-annotator calibration is essential and remains future work.

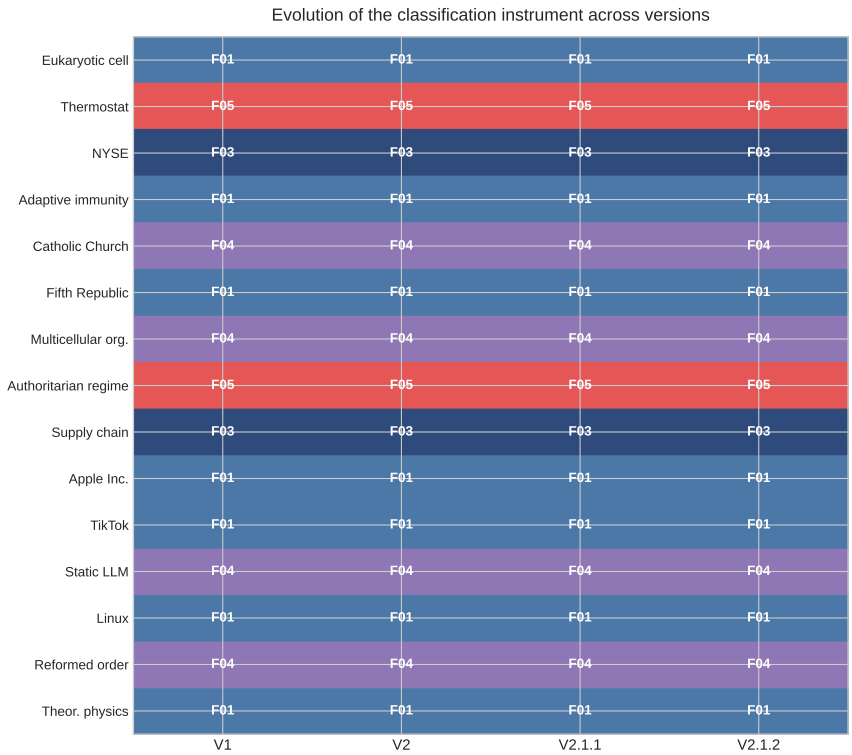

Figure 4: Evolution of regime assignments across four classifier versions. Classification stability across instrument revisions supports the structural robustness of the regime definitions.

2. **Small case set.** Fifteen systems provide a proof of concept, not a validation. The sampling design (maximal contrast) is appropriate for framework exploration but does not support statistical inference. Some regime zones remain unpopulated (F08, strict F06).

3. **Empirical thresholds.** Regime boundaries (e.g., $\Delta_{23} = 0.18$ for F01, $\Delta_{45} = 0.40$ for F05) were determined empirically during classifier development, not derived from theoretical principles. They are stable under scoring changes (V2.1 $\rightarrow$ V2.1.2) but not proven to be optimal.

4. **No temporal dynamics.** Regimes are defined as regions of a static property space. The transition diagram (F01 $\rightarrow$ F03 $\rightarrow$ F06 $\rightarrow$ F01) is a structural possibility map, not a dynamical model.

5. **Missing relational axes.** The five axes capture internal organizational properties but not inter-system relations. Transmissibility (cross-substrate portability) and environmental openness are identified candidates for future expansion.

8 Conclusion and Outlook

This work introduced an exploratory framework for representing heterogeneous complex systems within a shared organizational property space. Five axes describe structural properties (hierarchical depth, propagation, integration, normativity, revision), and seven regimes are defined as constraint-bounded regions primarily determined by gradient relationships between these axes.

The central empirical finding is that gradient contrasts — particularly Δ_{23} (propagation–integration) and Δ_{45} (normativity–revision) — are more discriminant for regime identification than absolute axis values. This gradient primacy was confirmed by the invariance diagnostic (97.5% for Δ_{23}) and by the failure of absolute-threshold classification in V1.

The classifier passes logical consistency and non-regression tests. Known weaknesses — the soft F01/F06 frontier, dead zones in the unit cube, and marginal invariance failures for T_2/T_3 — are documented and structurally understood.

Three directions for future work appear particularly important. First, inter-annotator calibration using the cumulative sub-criteria protocol. Second, empirical expansion targeting unpopulated regime zones (F08, strict F06) and testing on 30+ systems. Third, exploration of formal proxies for A_3 (mutual information, following Mediano et al., 2022) and A_2 (network connectivity metrics) to bridge qualitative scoring and quantitative measurement.

More broadly, the framework suggests that comparative analysis of heterogeneous systems may benefit from focusing on *relationships between organizational properties* rather than on absolute property levels alone. The complete classifier code, scoring protocol, and diagnostic scripts are publicly available to support replication and extension.[2]

9 Appendix A. Scoring Protocol

9.1 A.1 General principle

Each axis is operationalized through five sub-criteria scored 0 (absent), 0.5 (partial), or 1 (clearly present). The axis score is the mean:

$$A_i = \frac{1}{5}\sum_{k=1}^{5} c_{ik}$$

9.2 A.2 A1 — Hierarchical depth

- H1: ≥ 2 distinct causal levels
- H2: ≥ 3 distinct causal levels
- H3: ≥ 4 distinct causal levels
- H4: Functionally differentiated levels
- H5: Bidirectional causality between levels

9.3 A.3 A2 — Propagation capacity (V2.1.1: scale-invariant formulation)

- P1: Local perturbation affects at least one other functional module
- P2: Propagation traverses at least one hierarchical level
- P3: Propagation modifies the observable global state
- P4: Isolation is difficult without structural modification
- P5: Non-trivial functional coupling

[2] https://github.com/MMli97/1-morphodynamic-index

9.4 A.4 A3 — Integration

- I1: Explicit coordination mechanism exists
- I2: Observable variance reduction
- I3: Multi-level synchronization
- I4: Global feedback loops
- I5: Maintenance of coherent global state

9.5 A.5 A4 — Normativity

- N1: Dynamic attractor exists
- N2: Active error correction
- N3: Hierarchy of regulatory priorities
- N4: Internal stabilization mechanism
- N5: Resistance to prolonged perturbation

9.6 A.6 A5 — Revision capacity (V2.1.1: endogenous plasticity formulation)

- R1: Local parametric adjustment
- R2: Durable internal configuration change
- R3: Network or structural reconfiguration
- R4: Modification of regulatory mechanisms
- R5: Capacity to produce new operating rules

10 Appendix B. Case Table

All 15 systems classified inside ($d = 0$). Six cases on documented overlap boundaries (margin = 0).

11 Appendix C. Diagnostic Summary

12 Appendix D. Version History

Version	Key change	Classification result
V1	Absolute thresholds	3/15 inside
V2	Gradient-primary; F09 added	15/15 inside
V2.1.1	A_2 scale-invariance; A_5 endogenous plasticity	Cell corrected (F05$→$F01)
V2.1.2	F01/F03 boundary: Δ_{23} 0.15$→$0.18 / 0.25$→$0.23	Grey zone halved; 15/15 stable

Table 1: Empirical case set — V2.1.2 classifier (central estimates).

System	A_1	A_2	A_3	A_4	A_5	Δ_{23}	Δ_{45}	Δ_{12}	Regime	2nd
Eukaryotic cell	0.70	0.60	0.75	0.65	0.35	−0.15	+0.30	+0.10	F01	F04
Thermostat	0.20	0.40	0.45	0.50	0.00	−0.05	+0.50	−0.20	F05	F01
Financial market (NYSE)	0.60	0.75	0.50	0.30	0.20	+0.25	+0.10	−0.15	F03	F01
Adaptive immune system	0.75	0.65	0.70	0.65	0.55	−0.05	+0.10	+0.10	F01	F06
Catholic Church	0.80	0.55	0.60	0.75	0.10	−0.05	+0.65	+0.25	F04	F05
Fifth Republic	0.65	0.60	0.55	0.60	0.40	+0.05	+0.20	+0.05	F01	F06
Multicellular organism	0.85	0.70	0.80	0.75	0.30	−0.10	+0.45	+0.15	F04	F05
Authoritarian regime	0.70	0.65	0.45	0.80	0.05	+0.20	+0.75	+0.05	F05	F04
Supply chain	0.55	0.80	0.45	0.35	0.15	+0.35	+0.20	−0.25	F03	F01
Apple Inc.	0.75	0.65	0.70	0.70	0.50	−0.05	+0.20	+0.10	F01	F06
TikTok	0.60	0.75	0.65	0.55	0.45	+0.10	+0.10	−0.15	F01	F06
Static LLM	0.90	0.55	0.75	0.05	0.00	−0.20	+0.05	+0.35	F04	F09
Linux ecosystem	0.70	0.55	0.70	0.60	0.55	−0.15	+0.05	+0.15	F01	F06
Reformed order	0.75	0.50	0.65	0.70	0.25	−0.15	+0.45	+0.25	F04	F05
Theoretical physics	0.70	0.55	0.65	0.55	0.60	−0.10	−0.05	+0.15	F01	F06

Table 2: Diagnostic results — V2.1.2 classifier, 100,000 sample points.

Test	Object	Result	Key metric
T1	Logical consistency	PASS	0 contradictions, 7/7 SAT
T2	Overlap / ambiguity	FAIL (localized)	$P(m < 0.01) = 19.7\%$, mainly F01/F06
T3	Coverage	FAIL (inflated)	$P(d_{\min} > 0.10) = 24.6\%$; 58.4% inside
T4	Noise stability	PARTIAL FAIL	5.6% per-point flip rate; 0.8% mean; boundary coherence 97.7%
T5	Gradient invariance	PARTIAL PASS	97.5% (Δ_{23}), 93.4% (Δ_{45}), 93.8% (Δ_{12})
T6	Non-regression	PASS	15/15 match, 15/15 inside

13 References

- Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.
- Bar-Yam, Y. (1997). *Dynamics of Complex Systems*. Addison-Wesley.
- Beer, S. (1972). *Brain of the Firm*. Allen Lane.
- Bengio, Y., Courville, A., & Vincent, P. (2013). Representation learning: A review and new perspectives. *IEEE Trans. Pattern Anal. Mach. Intell.*, 35(8), 1798–1828.
- Bertalanffy, L. von (1968). *General System Theory*. George Braziller.
- Boyd, S. & Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press.
- Canguilhem, G. (1966). *Le Normal et le Pathologique*. PUF.
- Chang, H. (2004). *Inventing Temperature*. Oxford University Press.
- Giere, R. N. (1988). *Explaining Science: A Cognitive Approach*. University of Chicago Press.
- Holling, C. S. (1973). Resilience and stability of ecological systems. *Annual Rev. Ecol. Syst.*, 4, 1–23.
- Holling, C. S. (2001). Understanding the complexity of economic, ecological, and social systems. *Ecosystems*, 4(5), 390–405.
- Kauffman, S. (1993). *The Origins of Order*. Oxford University Press.
- Ladyman, J., Lambert, J., & Wiesner, K. (2013). What is a complex system? *Euro. J. Phil. Sci.*, 3, 33–67.
- Maturana, H. R. & Varela, F. J. (1980). *Autopoiesis and Cognition*. Reidel.
- Mediano, P. A. M., Rosas, F. E., et al. (2022). Greater than the parts: A review of the information decomposition approach to causal emergence. *Phil. Trans. R. Soc. A*, 380, 20210246.
- Simondon, G. (1958/2005). *L'Individuation à la lumière des notions de forme et d'information*. Millon.
- Thom, R. (1972). *Stabilité structurelle et morphogenèse*. Benjamin.
- Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5, 42.
- Tononi, G. & Edelman, G. M. (1998). Consciousness and complexity. *Science*, 282, 1846–1851.
- van Fraassen, B. (1980). *The Scientific Image*. Oxford University Press.
- van Fraassen, B. (2008). *Scientific Representation*. Oxford University Press.
- Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind*. MIT Press.
- Walker, B. & Salt, D. (2006). *Resilience Thinking*. Island Press.